185997: granitic gneiss, Tandys Bore

(Cundimurra Monzogranite, Tuckanarra Suite, Murchison Domain, Youanmi Terrane, Yilgarn Craton)

Location and sampling

Ninghan (SH 50-7), Bungar (2539)
MGA Zone 50, 610848E 6785494N

Sampled on 19 November 2008

This sample was collected from the north slope of a large granite hill on Wydgee Station, about 8.2 km northwest of New Well, 6.0 km east of Risdens Well, and 0.6 km north of Tandys Bore.

Tectonic unit/relations

The unit sampled is the Cundimurra Monzogranite of the 2690–2665 Ma Tuckanarra Suite (Van Kranendonk et al., 2013). Plutons of the Tuckanarra Suite consist of strongly foliated and locally magmatically layered granodiorite to monzogranite. Tuckanarra Suite granites intrude the Murchison Supergroup and are crosscut by the undeformed granitic plutons of the Bald Rock Supersuite (Van Kranendonk et al. 2013). A sample of the Cundimurra Monzogranite, sampled 15.3 km to the south, yielded a crystallization age of 2681 ± 5 Ma (GSWA 198212; Wingate et al., 2015). About 60 km further north along the Cundimurra Shear Zone, Wang (1998) dated a metamonzogranite at 2688 ± 6 Ma.

Petrographic description

The sample is a weakly gneissose, garnet-bearing aplite or microgranite gneiss, composed of about 50–60% quartz, 30–40% plagioclase, 10% K-feldspar, and accessory biotite, garnet, chlorite, sericite, iron oxide minerals, and zircon. The rock displays segregated bands dominated by feldspar and bands dominated by quartz. Fine- to medium-grained anhedral plagioclase and crosshatch-twinned microcline occur within a microcrystalline to fine-grained mosaic of interlocked quartz grains. Quartz commonly forms elongated ribbons in a weakly aligned fabric. Plagioclase (oligoclase, An25-28) is partly clouded and displays albite twinning. Rare anhedral to subhedral, partly resorbed garnet may be of igneous origin or a relic of partly melted metamorphic rocks. The observed paragenetic sequence is garnet, plagioclase, microcline and quartz, biotite, chlorite, and sericite.

Zircon morphology

Zircons isolated from this sample are pale to dark brown or opaque, and anhedral to subhedral. The crystals are up to 150 µm long, and mainly equant, with aspect ratios up to 3:1. In cathodoluminescence (CL) images, most crystals display concentric zoning. A CL image of representative zircons is shown in Figure 1.

Analytical details

This sample was analysed on 18 April 2011, using SHRIMP-B. Three analyses of the Temora standard were obtained during the session, and indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.46% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Seven analyses were obtained from six zircons. Only 10 zircons were recovered from this sample. Results are listed in Table 1, and shown in a concordia diagram (Fig. 2).

Interpretation

The analyses are essentially concordant (Fig. 2), and can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises two analyses of one zircon (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2679 ± 8 Ma (MSWD = 1.0).

Group X comprises five analyses (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2953–2713 Ma.

The date of 2679 ± 8 Ma for the two analyses in Group I is tentatively interpreted as the magmatic crystallization age of the granite. Alternatively, this zircon could be inherited, in which case the date of 2679 ± 8 Ma represents a maximum age for granite crystallization.

The dates of 2953–2713 Ma for the five analyses in Group X are interpreted as the ages of inherited components.

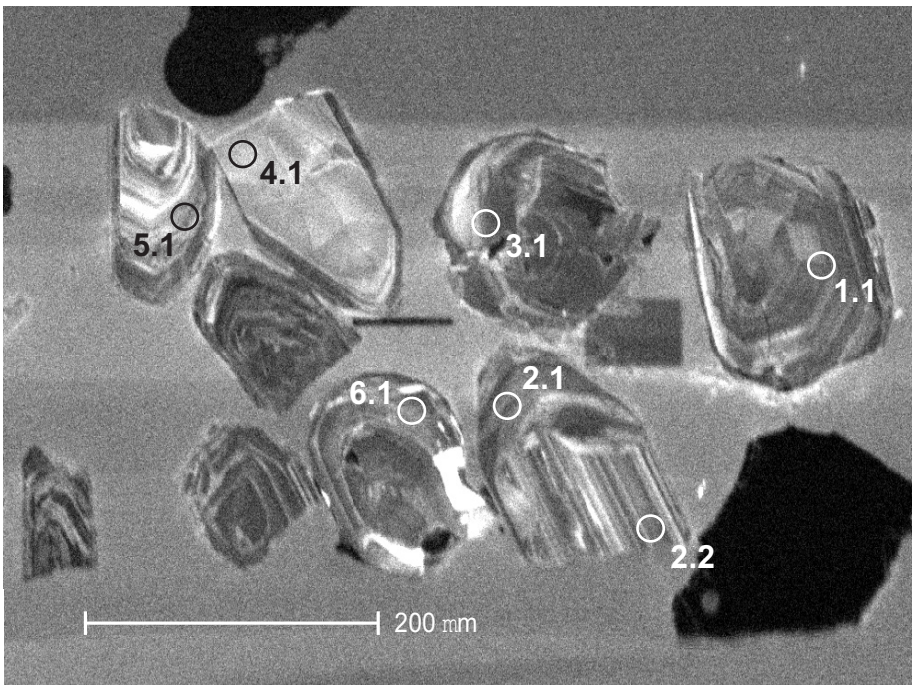

Figure 1. Cathodoluminescence image of representative zircons from sample 185997: granitic gneiss, Tandys Bore. Numbered circles indicate the approximate locations of analysis sites.

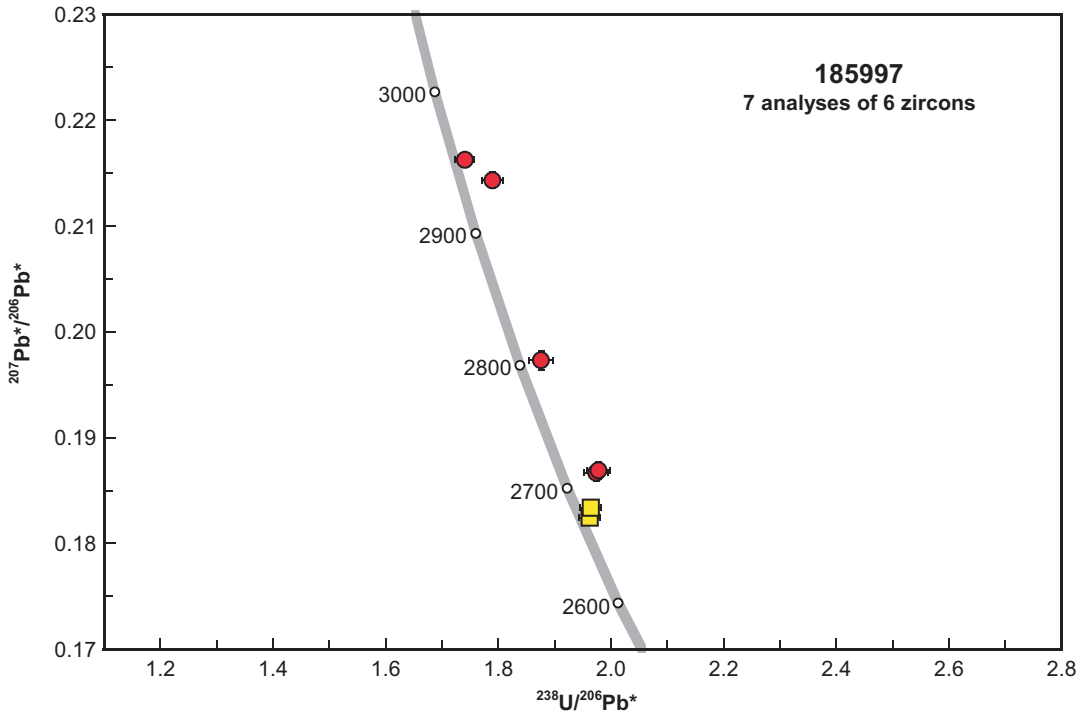

Figure 2. U–Pb analytical data for sample 185997: granitic gneiss, Tandys Bore. Yellow squares indicate Group I (magmatic zircons); red circles indicate Group X (xenocrystic zircons).

Table 1. Ion microprobe analytical results for zircons from sample 185997: granitic gneiss, Tandys Bore

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$\frac{^{232}Th}{^{238}U}$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	2	2.1	231	21	0.09	0.109	1.960	0.018	0.18342	0.00056	1.962	0.018	0.18244	0.00060	2655	20	2675	5	0.8
I	7	2.2	227	53	0.24	0.039	1.963	0.019	0.18371	0.00069	1.964	0.019	0.18337	0.00070	2653	21	2684	6	1.1
X	4	4.1	132	25	0.19	0.093	1.972	0.021	0.18749	0.00073	1.974	0.021	0.18666	0.00077	2642	23	2713	7	2.6
X	5	5.1	158	41	0.27	0.140	1.976	0.020	0.18812	0.00075	1.978	0.020	0.18688	0.00080	2637	23	2715	7	2.9
X	3	3.1	108	39	0.38	0.126	1.874	0.021	0.19840	0.00085	1.876	0.021	0.19728	0.00091	2754	26	2804	8	1.8
X	6	6.1	176	63	0.37	0.126	1.788	0.018	0.21542	0.00071	1.790	0.018	0.21430	0.00074	2861	23	2938	6	2.6
X	1	1.1	213	100	0.48	0.118	1.739	0.017	0.21729	0.00065	1.741	0.017	0.21624	0.00068	2925	23	2953	5	0.9

References

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Recommended reference for this publication

Wingate, MTD, Kirkland, CL and Ivanic, TJ 2015, 185997: granitic gneiss, Tandys Bore; Geochronology Record 1229: Geological Survey of Western Australia, 3p.

Data obtained: 18 April 2011

Data released: 28 February 2015